Predictive AI vs. Generative AI

What will happen? vs. **What could this look like?**

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Reference and content - IBM think series

Pop quiz: predictive or generative?

1



Your bank blocks a suspicious card payment

2



A chatbot drafts your email for you

3



An airline estimates how many seats will fill

4



An AI assistant writes an explanation and suggests corrected code

5



Your inbox sorts mail into spam or not spam

6



An AI makes a poster from a text prompt

Hint: is it forecasting something, or creating something?

Two questions, two kinds of AI

P

PREDICTIVE AI

What will happen?

Forecasts a measurable outcome from historical data.

G

GENERATIVE AI

What could this look like?

Creates new content that resembles its training data.



Your bank blocks a suspicious card payment #1



A chatbot drafts your email for you #2



An airline estimates how many seats will fill #3



An AI assistant writes an explanation and suggests corrected code #4



Your inbox sorts mail into spam or not spam #5



An AI makes a poster from a text prompt #6

Side by side: output, data, behavior

	Predictive AI	Generative AI
Output	Numbers, categories or probabilities (hypothetical example: an 87% chance a customer cancels)	Content: text, images or code
How is it judged?	Evaluate predictions against observed outcomes	Many valid answers, so people or criteria judge quality
Input data	Typically structured: rows, columns, sensor readings	Typically unstructured: billions of words, photos, code snippets
Behavior	Predictive models often repeat the same prediction	Generative responses can vary with sampling settings
Typical example	Will this transaction be fraud?	Write me an email

Common patterns, not strict rules: both can use structured and unstructured data.

Predictive AI: three flavors



Regression

Predicts a number

Learns the relationship between input variables and a continuous numerical output.

EXAMPLE

Units you'll sell next month. The price a house will fetch.



Classification

Predicts a category

The output is discrete: it's this, or it's that.

EXAMPLE

Spam or not spam. Fraud or not fraud.



Time series

Predicts change over time

Pays attention to patterns like seasonality and trends.

EXAMPLE

Stock prices. Server load. Electricity demand.

Predictive AI: what's under the hood

For tables of data



Decision trees

A flowchart of yes/no questions that ends in a prediction.



Random forests

A bunch of decision trees voting together.



Gradient boosting

Builds trees one after another, each correcting the errors of the last.

For time series



ARIMA / SARIMA

Autoregressive Integrated moving average

Classic statistical models for trends and seasonality.



LSTMs

Long-short term memory

Deep learning that remembers patterns over long stretches of time.

Generative AI: how it creates

TEXT: TRANSFORMERS AND ATTENTION

For text, a common architecture is the transformer. Its attention mechanism helps the model use relevant context.

Transformers power large language models.

Attention lets the model weigh which parts of the input matter for each word it writes.

Massive training data teaches it patterns well enough to create new content.

ATTENTION, ILLUSTRATED

The	cat	chased	the	mouse
because	it	was	hungry	

To read “it”, the model weighs “cat” more than “mouse”. (Illustration)

IMAGES: DIFFUSION MODELS

Training: noise is added to real images, and the model learns to reverse it.

Generating: start from pure noise and remove it step by step, guided by your prompt.



Pure noise

New image

Remove noise, step by step

So where do LLMs fit?



The mechanism

Inside, an LLM is a next-token prediction engine. It predicts one small piece of text at a time.



The analogy

A gasoline car is 'a series of controlled explosions'. True, but not useful for understanding what a car does.



The purpose

LLMs are built to produce new text, code and answers. Their purpose is primarily generative.

Verdict: we file LLMs under generative AI. Judge a tool by what it is for, not only by how it works inside.

Where each one is used

Predictive AI



Fraud detection
Fraud or not fraud, on every card swipe



Demand forecasting
Inventory to stock, seats to fill



Predictive maintenance
Replace a part right before it fails



Credit scoring
Probability someone will default

Generative AI



Content creation
Emails, marketing copy, posts



Code assistance
Write, explain and debug code



Conversational AI
Customer service bots that chat



Summarization
Long documents, distilled

Better together

Example 1: find the problem, then respond to it

PREDICTIVE MODEL

Which customers are likely to churn?

→

THE HAND-OFF

A list of at-risk customers

→

GENERATIVE MODEL

Writes a retention email for each one, using an approved offer

Example 2: help a predictive model when data is hard to get

THE PROBLEM

Real data is scarce or sensitive

→

GENERATIVE AI

Creates synthetic training data

→

PREDICTIVE MODEL

Trains on that data to make predictions

Remember two questions

P

PREDICTIVE AI

What will happen?

Numbers, categories or probabilities you can evaluate.

G

GENERATIVE AI

What could this look like?

New content with many valid answers.

Best together: prediction finds the problem, generation crafts the response.

Thank you! Questions?

Predictive AI is the fortune teller.

Generative AI is the storyteller.

Great teams know which one to ask.